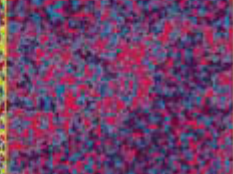
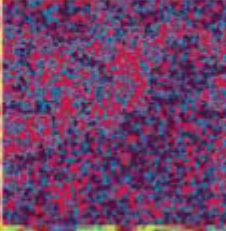
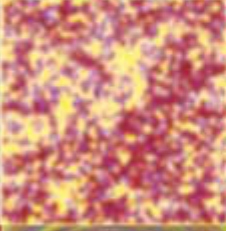
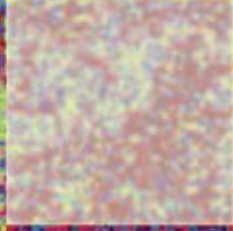
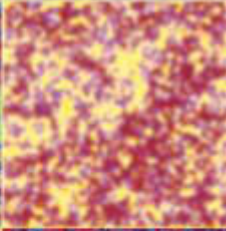
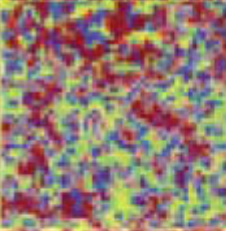
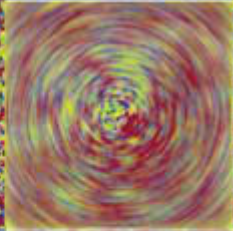
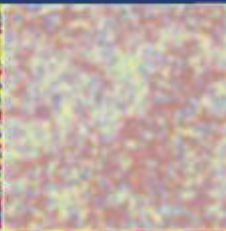
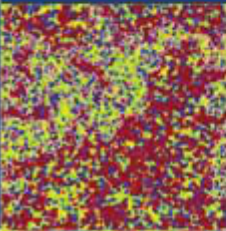
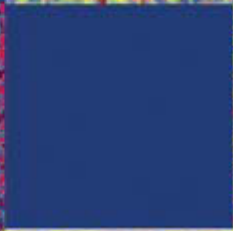
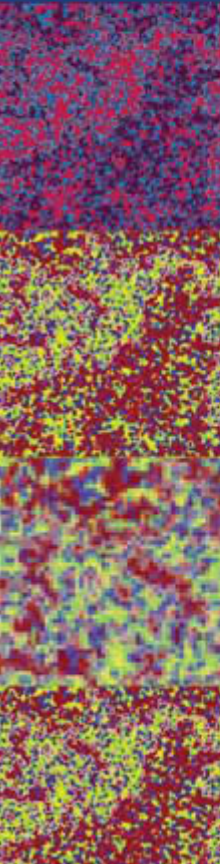
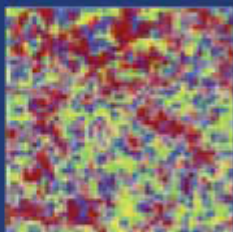


Computational Physics

J. M. Thijssen

Second edition



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Computational Physics

Second Edition

This Second Edition has been fully updated. The wide range of topics covered in the First Edition has been extended with new chapters on finite element methods and lattice Boltzmann simulation. New sections have been added to the chapters on density functional theory, quantum molecular dynamics, Monte Carlo simulation and diagonalisation of one-dimensional quantum systems.

The book covers many different areas of physics research and different computational methodologies, with an emphasis on condensed matter physics and physical chemistry. It includes computational methods such as Monte Carlo and molecular dynamics, various electronic structure methodologies, methods for solving partial differential equations, and lattice gauge theory. Throughout the book, the relations between the methods used in different fields of physics are emphasised. Several new programs are described and these can be downloaded from www.cambridge.org/9780521833462

The book requires a background in elementary programming, numerical analysis and field theory, as well as undergraduate knowledge of condensed matter theory and statistical physics. It will be of interest to graduate students and researchers in theoretical, computational and experimental physics.

JOS THIJSEN is a lecturer at the Kavli Institute of Nanoscience at Delft University of Technology.

COMPUTATIONAL PHYSICS

Second Edition

JOS THIJSEN

Kavli Institute of Nanoscience, Delft University of Technology



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Contents

Preface to the first edition	<i>page</i> xi
Preface to the second edition	xiv
1 Introduction	1
1.1 Physics and computational physics	1
1.2 Classical mechanics and statistical mechanics	1
1.3 Stochastic simulations	4
1.4 Electrodynamics and hydrodynamics	5
1.5 Quantum mechanics	6
1.6 Relations between quantum mechanics and classical statistical physics	7
1.7 Quantum molecular dynamics	8
1.8 Quantum field theory	9
1.9 About this book	9
Exercises	11
References	13
2 Quantum scattering with a spherically symmetric potential	14
2.1 Introduction	14
2.2 A program for calculating cross sections	18
2.3 Calculation of scattering cross sections	25
Exercises	27
References	28
3 The variational method for the Schrödinger equation	29
3.1 Variational calculus	29
3.2 Examples of variational calculations	32
3.3 Solution of the generalised eigenvalue problem	36
3.4 Perturbation theory and variational calculus	37

Exercises	39
References	41
4 The Hartree–Fock method	43
4.1 Introduction	43
4.2 The Born–Oppenheimer approximation and the independent-particle method	44
4.3 The helium atom	46
4.4 Many-electron systems and the Slater determinant	52
4.5 Self-consistency and exchange: Hartree–Fock theory	54
4.6 Basis functions	60
4.7 The structure of a Hartree–Fock computer program	69
4.8 Integrals involving Gaussian functions	73
4.9 Applications and results	77
4.10 Improving upon the Hartree–Fock approximation	78
Exercises	80
References	87
5 Density functional theory	89
5.1 Introduction	89
5.2 The local density approximation	95
5.3 Exchange and correlation: a closer look	97
5.4 Beyond DFT: one- and two-particle excitations	101
5.5 A density functional program for the helium atom	109
5.6 Applications and results	114
Exercises	116
References	119
6 Solving the Schrödinger equation in periodic solids	122
6.1 Introduction: definitions	123
6.2 Band structures and Bloch’s theorem	124
6.3 Approximations	126
6.4 Band structure methods and basis functions	133
6.5 Augmented plane wave methods	135
6.6 The linearised APW (LAPW) method	141
6.7 The pseudopotential method	144
6.8 Extracting information from band structures	160
6.9 Some additional remarks	162
6.10 Other band methods	163

Exercises	163
References	167
7 Classical equilibrium statistical mechanics	169
7.1 Basic theory	169
7.2 Examples of statistical models; phase transitions	176
7.3 Phase transitions	184
7.4 Determination of averages in simulations	192
Exercises	194
References	195
8 Molecular dynamics simulations	197
8.1 Introduction	197
8.2 Molecular dynamics at constant energy	200
8.3 A molecular dynamics simulation program for argon	208
8.4 Integration methods: symplectic integrators	211
8.5 Molecular dynamics methods for different ensembles	223
8.6 Molecular systems	232
8.7 Long-range interactions	241
8.8 Langevin dynamics simulation	247
8.9 Dynamical quantities: nonequilibrium molecular dynamics	251
Exercises	253
References	259
9 Quantum molecular dynamics	263
9.1 Introduction	263
9.2 The molecular dynamics method	266
9.3 An example: quantum molecular dynamics for the hydrogen molecule	272
9.4 Orthonormalisation; conjugate gradient and RM-DIIS techniques	278
9.5 Implementation of the Car–Parrinello technique for pseudopotential DFT	289
Exercises	290
References	293
10 The Monte Carlo method	295
10.1 Introduction	295
10.2 Monte Carlo integration	296
10.3 Importance sampling through Markov chains	299

10.4	Other ensembles	310
10.5	Estimation of free energy and chemical potential	316
10.6	Further applications and Monte Carlo methods	319
10.7	The temperature of a finite system	330
	Exercises	334
	References	335
11	Transfer matrix and diagonalisation of spin chains	338
11.1	Introduction	338
11.2	The one-dimensional Ising model and the transfer matrix	339
11.3	Two-dimensional spin models	343
11.4	More complicated models	347
11.5	‘Exact’ diagonalisation of quantum chains	349
11.6	Quantum renormalisation in real space	355
11.7	The density matrix renormalisation group method	358
	Exercises	365
	References	370
12	Quantum Monte Carlo methods	372
12.1	Introduction	372
12.2	The variational Monte Carlo method	373
12.3	Diffusion Monte Carlo	387
12.4	Path-integral Monte Carlo	398
12.5	Quantum Monte Carlo on a lattice	410
12.6	The Monte Carlo transfer matrix method	414
	Exercises	417
	References	421
13	The finite element method for partial differential equations	423
13.1	Introduction	423
13.2	The Poisson equation	424
13.3	Linear elasticity	429
13.4	Error estimators	434
13.5	Local refinement	436
13.6	Dynamical finite element method	439
13.7	Concurrent coupling of length scales: FEM and MD	440
	Exercises	445
	References	446

14 The lattice Boltzmann method for fluid dynamics	448
14.1 Introduction	448
14.2 Derivation of the Navier–Stokes equations	449
14.3 The lattice Boltzmann model	455
14.4 Additional remarks	458
14.5 Derivation of the Navier–Stokes equation from the lattice Boltzmann model	460
Exercises	463
References	464
15 Computational methods for lattice field theories	466
15.1 Introduction	466
15.2 Quantum field theory	467
15.3 Interacting fields and renormalisation	473
15.4 Algorithms for lattice field theories	477
15.5 Reducing critical slowing down	491
15.6 Comparison of algorithms for scalar field theory	509
15.7 Gauge field theories	510
Exercises	532
References	536
16 High performance computing and parallelism	540
16.1 Introduction	540
16.2 Pipelining	541
16.3 Parallelism	545
16.4 Parallel algorithms for molecular dynamics	552
References	556
Appendix A Numerical methods	557
A1 About numerical methods	557
A2 Iterative procedures for special functions	558
A3 Finding the root of a function	559
A4 Finding the optimum of a function	560
A5 Discretisation	565
A6 Numerical quadratures	566
A7 Differential equations	568
A8 Linear algebra problems	590
A9 The fast Fourier transform	598
Exercises	601
References	603

Appendix B	Random number generators	605
B1	Random numbers and pseudo-random numbers	605
B2	Random number generators and properties of pseudo-random numbers	606
B3	Nonuniform random number generators	609
	Exercises	611
	References	612
Index		613

Preface to the first edition

This is a book on computational methods used in theoretical physics research, with an emphasis on condensed matter applications.

Computational physics is concerned with performing computer calculations and simulations for solving physical problems. Although computer memory and processor performance have increased dramatically over the last two decades, most physical problems are too complicated to be solved without approximations to the physics, quite apart from the approximations inherent in any numerical method. Therefore, most calculations done in computational physics involve some degree of approximation. In this book, emphasis is on the derivation of algorithms and the implementation of these: it is a book which tells you how methods work, why they work, and what the approximations are. It does not contain extensive discussions on results obtained for large classes of different physical systems.

This book is not elementary: the reader should have a background in basic undergraduate physics and in computing. Some background in numerical analysis is also helpful. On the other hand, the topics discussed are not treated in a comprehensive way; rather, this book hopefully bridges the gap between more elementary texts by Koonin, Gould and Giordano, and specialised monographs and review papers on the applications described. The fact that a wide range of topics is included has the advantage that the many similarities in the methods used in seemingly very different fields could be highlighted. Many important topics and applications are however not considered in this book – the material presented obviously reflects my own expertise and interest.

I hope that this book will be useful as a source for intermediate and advanced courses on the subject. I furthermore hope that it will be helpful for graduates and researchers who want to increase their knowledge of the field.

Some variation in the degree of difficulty is inherent to the topics addressed in this book. For example, in molecular dynamics, the equations of motion of a collection of particles are solved numerically, and as such it is a rather elementary subject. However, a careful analysis of the integration algorithms used, the problem of performing these simulations in different statistical ensembles, and the problem of

treating long range forces with periodic boundary conditions, are much more difficult. Therefore, sections addressing advanced material are marked with an asterisk (*) – they can be skipped at first reading. Also, extensive theoretical derivations are sometimes moved to sections with asterisks, so that the reader who wants to write programs rather than go into the theory may use the results, taking the derivations for granted.

Aside from theoretical sections, implementations of algorithms are discussed, often in a step-by-step fashion, so that the reader can program the algorithms him- or herself. Suggestions for checking the program are included. In the exercises after each chapter, additional suggestions for programs are given, but there are also exercises in which the computer is not used. The computer exercises are marked by the symbol [C]; if the exercise is divided up into parts, this sign occurs before the parts in which a computer program is to be written (a problem marked with [C] may contain major parts which are to be done analytically). The programs are not easy to write – most of them took me a long time to complete! Some data-files and numerical routines can be found on www.cambridge.org/9780521833469.

The first person who suggested that I should write this book was Aloysio Janner. Thanks to the support and enthusiasm of my colleague and friend John Inglesfield in Nijmegen, I then started writing a proposal containing a draft of the first hundred pages. After we both moved to the University of Cardiff (UK), he also checked many chapters with painstaking precision, correcting the numerous errors, both in the physics and in the English; without his support, this book would probably never have been completed.

Bill Smith, from Daresbury Laboratories (UK), has checked the chapters on classical many-particle systems and Professor Konrad Singer those on quantum simulation methods. Simon Hands from the University of Swansea (UK) has read the chapter on lattice field theories, and Hubert Knops (University of Nijmegen, The Netherlands) those on statistical mechanics and transfer matrix calculations. Maziar Nekovee (Imperial College, London, UK) commented on the chapter on quantum Monte Carlo methods. I am very grateful for the numerous suggestions and corrections from them all. I am also indebted to Paul Hayman for helping me correcting the final version of the manuscript. Many errors in the book have been pointed out to me by colleagues and students. I thank Professor Ron Cohen in particular for spotting many mistakes and discussing several issues via email.

In writing this book, I have discovered that the acknowledgements to the author's family, often expressed in an apologetic tone as a result of the disruption caused by the writing process to family life, are too real to be disqualified as a cliché. My sons Maurice, Boudewijn and Arthur have in turn disrupted the process of writing in the most pleasant way possible, regularly asking me to show growing trees or fireworks on the screen of my PC, instead of the dull black-on-white text

windows. Boudewijn and Maurice's professional imitation of their dad, tapping on the keyboard, and sideways reading formulae, is promising for the future.

It is to my wife Ellen that I dedicate this book, with gratitude for her patience, strength and everlasting support during the long, and sometimes difficult time in which the book came into being.

Preface to the second edition

Six years have passed since the first edition of this book appeared. In these years I have learned a lot more about computational physics – a process which will hopefully never stop. I learned from books and papers, but also from the excellent colleagues with whom I worked on teaching and research during this period. Some of this knowledge has found its place in this edition, which is a substantial extension of the first.

New topics include finite elements, lattice Boltzmann simulation and density matrix renormalisation group, and there are quite a few sections here and there in the book which either give a more in-depth treatment of the material than can be found in the first edition, or extensions to widen the view on the subject matter. Moreover I have tried to eliminate as many errors as possible, but I am afraid that it is difficult for me to beat the entropy of possible things which can go wrong in writing a book of over 650 pages.

In Delft, where I have now a position involving a substantial amount of teaching, I worked for several years in the computational physics group of Simon the Leeuw. I participated in an exciting and enjoyable effort: teaching in an international context. Together with Rajiv Kalia, from Louisiana State, we let students from Delft collaborate with Louisiana students, having them do projects in the field of computational physics. Both Simon and Rajiv are experts in the field of molecular dynamics, and I learned a lot from them. Moreover, dealing with students and their questions has often forced me to deepen my knowledge in this field. Similar courses with Hiroshi Iyetomi from Niigata University in Japan, and now with Phil Duxbury at Michigan State have followed, and form my most enjoyable teaching experience. Much of the knowledge picked up in these courses has gone into the new material in this edition.

For one of the new parts of the book, the self-consistent pseudopotential and the Car–Parrinello program, I worked closely together with Erwin de Wolff for a few months. I am grateful for his support in this, and not least for his structured, neat way of tackling the problem.

Many students, university lecturers and researchers have shared their corrections on the text with me. I want to thank Ronald Cohen, Dominic Holland, Ari Harju,

John Mauro, Joachim Stolze and all the others whose names may have disappeared from my hard disks when moving to a new machine.

Preparing this edition in addition to the regular duties of a university position has turned out to be a demanding job, which has prevented me now and then from being a good husband and father. I thank Ellen and my sons Maurice, Boudewijn and Arthur for their patience and support, and express the hope that I will have more time for them in the future.

1

Introduction

1.1 Physics and computational physics

Solving a physical problem often amounts to solving an ordinary or partial differential equation. This is the case in classical mechanics, electrodynamics, quantum mechanics, fluid dynamics and so on. In statistical physics we must calculate sums or integrals over large numbers of degrees of freedom. Whatever type of problem we attack, it is very rare that analytical solutions are possible. In most cases we therefore resort to numerical calculations to obtain useful results. Computer performance has increased dramatically over the last few decades (see also [Chapter 16](#)) and we can solve complicated equations and evaluate large integrals in a reasonable amount of time.

Often we can apply numerical routines (found in software libraries for example) directly to the physical equations and obtain a solution. We shall see, however, that although computers have become very powerful, they are still unable to provide a solution to most problems without approximations to the physical equations. In this book, we shall focus on these approximations: that is, we shall concentrate on the development of computational methods (and also on their implementation into computer programs). In this introductory chapter we give a bird's-eye perspective of different fields of physics and the computational methods used to solve problems in these areas. We give examples of direct application of numerical methods but we also give brief and heuristic descriptions of the additional theoretical analysis and approximations necessary to obtain workable methods for more complicated problems which are described in more detail in the remainder of this book. The order adopted in the following sections differs somewhat from the order in which the material is treated in this book.

1.2 Classical mechanics and statistical mechanics

The motion of a point particle in one dimension subject to a force F depending on the particle's position x , and perhaps on the velocity $\dot{x}$ and on time t , is determined

by Newton's equation of motion:

$$m\ddot{x}(t) = F[x(t), \dot{x}(t), t]. \quad (1.1)$$

The (double) dot denotes a (double) derivative with respect to time. A solution can be found for each set of initial conditions $x(t_0)$ and $\dot{x}(t_0)$ given at some time t_0 . Analytical solutions exist for constant force, for the harmonic oscillator ($F = \kappa x^2/2$), and for a number of other cases. In [Appendix A7.1](#) a simple numerical method for solving this equation is described and this can be applied straightforwardly to arbitrary forces and initial conditions.

Interesting and sometimes surprising physical phenomena can now be studied. As an example, consider the Duffing oscillator [\[1\]](#), with a force given by

$$F[x, \dot{x}, t] = -\gamma \dot{x} + 2ax - 4bx^3 + F_0 \cos(\omega t). \quad (1.2)$$

The first term on the right hand side represents a velocity-dependent friction; the second and third terms are the force a particle feels when it moves in a double potential well $bx^4 - ax^2$, and the last term is an external periodic force. An experimental realisation is a pendulum consisting of an iron ball suspended by a thin string, with two magnets below it. The pendulum and the magnets are placed on a table which is moved back and forth with frequency ω . The string and the air provide the frictional force, the two magnets together with gravity form some kind of double potential well, and, in the reference frame in which the pendulum is at rest, the periodic motion of the table is felt as a periodic force. It turns out that the Duffing oscillator exhibits *chaotic behaviour* for particular values of the parameters γ , a , b , F_0 and ω . This means that the motion itself looks irregular and that a very small change in the initial conditions will grow and result in a completely different motion. [Figure 1.1](#) shows the behaviour of the Duffing oscillator for two nearly equal initial conditions, showing the sensitivity to these conditions. Over the past few decades, chaotic systems have been studied extensively. A system that often behaves chaotically is the weather: the difficulty in predicting the evolution of chaotic systems causes weather forecasts to be increasingly unreliable as they look further into the future, and occasionally to be dramatically wrong.

Another interesting problem is that of several particles, moving in three dimensions and subject to each other's gravitational interaction. Our Solar System is an example. For the simplest nontrivial case of three particles (for two particles, Newton has given the analytical solution), analytical solutions exist for particular configurations, but the general problem can only be solved numerically. This problem is called the *three-body problem* (N -body problem in general). The motion of satellites orbiting in space is calculated numerically using programs for the N -body problem, and the evolution of galaxies is calculated with similar programs using a large number of test particles (representing the stars). Millions of particles can

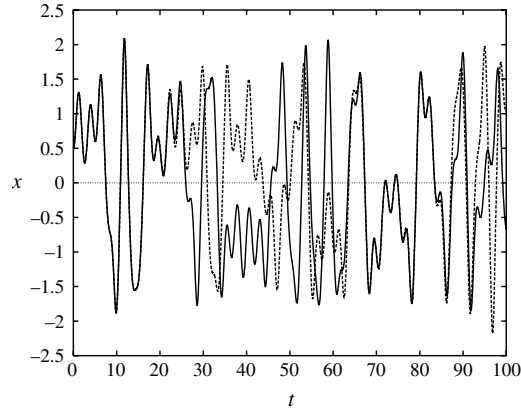


Figure 1.1. Solution of the Duffing oscillator. Parameters are $m = 1$, $a = 1/4$, $b = 1/2$, $F_0 = 2.0$, $\omega = 2.4$, $\gamma = 0.1$. Two solutions are shown: one with initial position $x_0 = 0.5$, the other with $x_0 = 0.5001$ ($\dot{x}_0 = 0$ in both cases). For these nearly equal initial conditions, the solutions soon become uncorrelated, showing the difficulty in predicting the time evolution of a chaotic system.

be treated using a combination of high-end computers and clever computational methods which will be considered in [Chapter 8](#). Electrostatic forces are related to gravitational forces, as both the gravitational and the electrostatic (Coulomb) potential have a $1/r$ form. The difference between the two is that electrostatic forces can be repulsive or attractive, whereas gravitational forces are always attractive.

Neutral atoms interact via a different potential: they attract each other weakly through induced polarisation, unless they come too close – then the Pauli principle causes the electron clouds to repel each other. The problem of many interacting atoms and molecules is a very important subfield of computational physics: it is called *molecular dynamics*. In molecular dynamics, the equations of motion for the particles are solved straightforwardly using numerical algorithms similar to those with which a Duffing oscillator is analysed, the main difference being the larger number of degrees of freedom in molecular dynamics. The aim of molecular dynamics simulations is to predict the behaviour of gases, liquids and solids (and systems in other phases, like liquid crystals). An important result is the equation of state: this is the relation between temperature, number of particles, pressure and volume. Also, the microscopic structure as exhibited by the pair correlation function, which is experimentally accessible via neutron scattering, is an interesting property which can be determined in simulations. There are, however, many problems and pitfalls associated with computer simulations: the systems that can be simulated are always much smaller than realistic systems, and simulating a system at a predefined temperature or chemical potential is nontrivial. All these aspects will be considered in [Chapter 8](#).

1.3 Stochastic simulations

In the previous section we have explained how numerical algorithms for solving Newton's equations of motion can be used to simulate liquids. The particles are moved around according to their mechanical trajectories which are governed by the forces they exert on each other. Another way of moving them around is to displace them in a random fashion. Of course this must be done in a controlled way, and not every move should be allowed, but we shall see in [Chapter 10](#) that it is possible to obtain information in this way similar to that obtained from molecular dynamics. This is an example of a *Monte Carlo* method – procedures in which random numbers play an essential role. The Monte Carlo method is not suitable for studying dynamical physical quantities such as transport coefficients, as it uses artificial dynamics to simulate many-particle systems.

Random number generators can also be used in *direct simulations*: some process of which we do not know the details is replaced by a random generator. If you simulate a card game, for example, the cards are distributed among the players by using random numbers. An example of a direct simulation in physics is diffusion limited aggregation (DLA), which describes the growth of dendritic clusters (see [Figure 1.2](#)). Consider a square lattice in two dimensions. The sites of the lattice are either occupied or unoccupied. Initially, only one site in the centre is occupied. We release a random walker from the boundary of the lattice. The walker moves over the lattice in a stepwise fashion. At each step, the walker moves from a site to

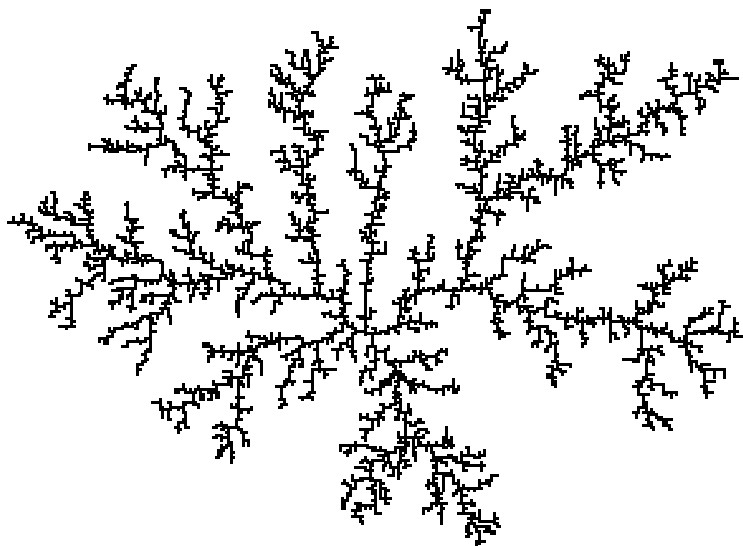


Figure 1.2. Dendritic cluster grown in a DLA simulation. The cluster consists of 9400 sites and it was grown on a 175×175 lattice.

one of its neighbour sites, which is chosen at random (there are four neighbours for each site in the interior of the lattice; the boundary sites have three neighbours, or two if they lie on a corner). If the walker arrives at a site neighbouring the occupied central site, it sticks there, so that a two-site cluster is formed. Then a new walker is released from the boundary. This walker also performs a random walk on the lattice until it arrives at a site neighbouring the cluster of two occupied sites, to form a three-site cluster, and so on. After a long time, a dendritic cluster is formed (see [Figure 1.2](#)), which shows a strong resemblance to actual dendrites formed in crystal growth, or by growing bacterial colonies [2], frost patterns on the window and numerous other physical phenomena.

This shows again that interesting physics can be studied by straightforward application of simple algorithms. In [Chapter 10](#) we shall concentrate on the Monte Carlo method for studying many-particle systems at a predefined temperature, volume and particle number. This technique is less direct than DLA, and, just as in molecular dynamics, studying the system for different predefined parameters, such as chemical potential, and evaluating free energies are nontrivial aspects which need further theoretical consideration. The Monte Carlo method also enables us to analyse lattice spin models, which are important for studying magnetism and field theory (see below). These models cannot always be analysed using molecular dynamics methods, and Monte Carlo is often the only tool we have at our disposal in that case. There also exist alternative, more powerful techniques for simulating dendrite formation, but these are not treated in this book.

1.4 Electrodynamics and hydrodynamics

The equations of electrodynamics and hydrodynamics are partial differential equations. There exist numerical methods for solving these equations, but the problem is intrinsically demanding because the fields are continuous and an infinite number of variables is involved. The standard approach is to apply some sort of *discretisation* and consider the solution for the electric potential or for the flow field only on the points of the discrete grid, thus reducing the infinite number of variables to a finite number. Another method of solution consists of writing the field as a linear combination of smooth functions, such as plane waves, and solving for the best values of the expansion coefficients.

There exist several methods for solving partial differential equations: finite difference methods (FDM), finite element methods (FEM), Fourier transform methods and multigrid methods. These methods are also very often used in engineering problems, and are essentially the domain of numerical analysis. The finite element method is very versatile and therefore receives our particular attention in [Chapter 13](#). The other methods can be found in [Appendix A7.2](#).

1.5 Quantum mechanics

In quantum mechanics we regularly need to solve the Schrödinger equation for one or more particles. There is usually an external potential felt by the particles, and in addition there might be interactions between the particles. For a single particle moving in one dimension, the stationary form of the Schrödinger equation reduces to an ordinary differential equation, and techniques similar to those used in solving Newton's equations can be used. The main difference is that the stationary Schrödinger equation is an eigenvalue equation, and in the case of a discrete spectrum, the energy eigenvalue must be varied until the wave function is physically acceptable, which means that it matches some boundary conditions and is normalisable. Examples of this direct approach are discussed in [Appendix A](#), in particular Problem A4.

In two and more dimensions, or if we have more than one particle, or if we want to solve the time-dependent Schrödinger equation, we must solve a partial differential equation. Sometimes, the particular geometry of the problem and the boundary conditions allow us to reduce the complexity of the problem and transform it into ordinary differential equations. This will be done in [Chapter 2](#), where we shall study particles scattering off a spherically symmetric potential.

Among the most important quantum problems in physics is the behaviour of electrons moving in the field generated by nuclei, which occurs in atoms, molecules and solids. This problem is treated quite extensively in this book, but the methods we develop for it are also applied in nuclear physics. Solving the Schrödinger equation for one electron moving in the potential generated by the atomic static nuclei is already a difficult problem, as it involves solving a partial differential equation. Moreover, the potential is strong close to the nuclei and weak elsewhere, so the typical length scale of the wave function varies strongly through space. Therefore, discretisation methods must use grids which are finer close to the nuclei, rendering such methods difficult. The method of choice is, in fact, to expand the wave function as a linear combination of fixed basis functions that vary strongly close to the nuclei and are smooth elsewhere, and find the optimal values for the expansion coefficients. This is an example of the *variational method*, which will be discussed in [Chapter 3](#). This application of the variational method leads to a matrix eigenvalue problem which can be solved very efficiently on a computer.

An extra complication arises when there are many (say N) electrons, interacting via the Coulomb potential, so that we must solve a partial differential equation in $3N$ dimensions. In addition to this we must realise that electrons are fermions and the many-electron wave function must therefore be antisymmetric with respect to exchange of any pair of electrons. Because of the large number of dimensions, solving the Schrödinger equation is not feasible using any of the

standard numerical methods for solving partial differential equations, so we must make approximations. One approach is the Hartree–Fock (HF) method, developed in the early days of quantum mechanics, which takes into account the antisymmetry of the many-electron wave function. This leads to an independent particle picture, in which each electron moves in the potential generated by the nuclei plus an average potential generated by the other electrons. The latter depends on the electronic wave functions, and hence the problem must be solved *self-consistently* – in [Chapter 4](#) we shall see how this is done. The HF method leads to wave functions that are fully antisymmetric, but contributions arising from the Coulomb interaction between the particles are taken into account in an approximate way, analogous to the way correlations are treated in the mean field approach in statistical mechanics.

Another approach to the quantum many-electron problem is given by *density functional theory* (DFT), which will be discussed in [Chapter 5](#). This theory, which is in principle exact, can in practice only be used in conjunction with approximate schemes to be discussed in [Chapter 5](#), the most important of which is the *local density approximation* (LDA). This also leads to an independent-particle Schrödinger equation, but in this case, the correlation effects resulting from the antisymmetry of the wave function are not incorporated exactly, leading to a small, unphysical interaction of an electron with itself (*self-interaction*). However, in contrast to Hartree–Fock, the approach does account (in an approximate way) for the dynamic correlation effects due to the electrons moving out of each other’s way as a result of the Coulomb repulsion between them.

All these approaches lead in the end to a matrix eigenvalue problem, whose size depends on the number of electrons present in the system. The resulting solutions enable us to calculate total energies and excitation spectra which can be compared with experimental results.

1.6 Relations between quantum mechanics and classical statistical physics

In the previous two sections we have seen that problems in classical statistical mechanics can be studied with Monte Carlo techniques, using random numbers, and that the solution of quantum mechanical problems reduces to solving matrix eigenvalue problems. It turns out that quantum mechanics and classical statistical mechanics are related in their mathematical structure. Consider for example the partition function for a classical mechanics system at temperature T , with degrees of freedom denoted by the variable X and described by an energy function (that is, a classical Hamiltonian) $\mathcal{H}$:

$$Z_{\text{Cl}} = \sum_X e^{-\mathcal{H}(X)/(k_{\text{B}}T)}, \quad (1.3)$$

and that of a quantum system with quantum Hamiltonian H :

$$Z_{\text{QM}} = \text{Tr}(e^{-H/(k_{\text{B}}T)}); \quad (1.4)$$

‘Tr’ denotes the trace of the operator following it. We will show in [Chapter 12](#) that in the path-integral formalism, the second expression can be transformed into the same form as the first one. Also, there is a strong similarity between the exponent occurring in the quantum partition function and the quantum time-evolution operator $U(t) = \exp(-itH/\hbar)$, so solving the time evolution of a quantum system is equivalent to evaluating a classical or quantum partition function, the difference being an imaginary factor $it/\hbar$ replacing the real factor $1/(k_{\text{B}}T)$, and taking the trace in the case of the quantum partition function rather than a sum over states in the classical analogue.

These mathematical analogies suggest that numerical methods for either classical statistical mechanics or quantum mechanics are applicable in both fields. Indeed, in [Chapter 11](#), we shall see that it is possible to analyse classical statistical spin problems on lattices by diagonalising large matrices. In [Chapter 12](#), on the other hand, we shall use Monte Carlo methods for solving quantum problems. These methods enable us to treat the quantum many-particle problem without systematic approximations, because, as will be shown in [Chapter 12](#), Monte Carlo techniques are very efficient for calculating integrals in many dimensions. This, as we have seen above, was precisely the problem arising in the solution of interacting many-particle systems.

1.7 Quantum molecular dynamics

Systems of many interacting atoms or molecules can be studied classically by solving Newton’s equations of motion, as is done in molecular dynamics. Pair potentials are often used to describe the atomic interactions, and these can be found from quantum mechanical calculations, using Hartree–Fock, density functional theory or quantum Monte Carlo methods. In a dense system, the pair potential is inadequate as the interactions between two particles in the system are influenced by other particles. In order to incorporate these effects in a simulation, it would be necessary to calculate the forces from full electronic structure calculations for all configurations occurring in the simulation. Car and Parrinello have devised a clever way to calculate these forces as the calculation proceeds, by combining density functional theory with molecular dynamics methods.

In the Car–Parrinello approach, electron correlations are not treated exactly because of the reliance on LDA (see [Section 1.5](#)), but it will be clear that it is an important improvement on fully classical simulations where the interatomic interactions are described by a simple form, such as pair potentials. It is possible

to include some damping mechanism in the equations of motion and then let the nuclei relax to their ground state positions, so that equilibrium configurations of molecules and solids can be determined (neglecting quantum fluctuations).

1.8 Quantum field theory

Quantum field theory provides a quantum description for fields: strings in one dimension, sheets in two dimensions, etc. Quantum field theory is also believed to describe elementary particles and their interactions. The best known example is quantum electrodynamics (QED) which gives a very accurate description of the interaction between charged spin-1/2 fermions (electrons) and electromagnetic fields. The results of QED are obtained using perturbation theory which works very well for this case, because the perturbative parameter remains small for all but the smallest length scales (at large length scales this is the fine structure constant).

In quantum chromodynamics (QCD), the theory which supposedly describes quarks bound together in a proton or neutron, the coupling constant grows large for large scales, and perturbation theory breaks down. One way to obtain useful results for this theory is to discretise space-time, and simulate the theory on this space-time lattice on a computer. This can be done using Monte Carlo or molecular dynamics techniques. The application of these techniques is far from easy as the QCD field theory is intrinsically complicated. A problem which needs to be addressed is efficiency, notably overcoming *critical slowing down*, which decreases the efficiency of simple Monte Carlo and molecular dynamics techniques for the cases which are of physical interest. The fact that quarks are fermions leads to additional complications.

QCD simulations relate quark masses to masses and interaction constants of hadrons (mesons, protons, neutrons).

1.9 About this book

In this book, the emphasis is on methods which do not merely involve straightforward application of numerical methods, and which are specific to problems studied in physics. In most cases, the theory is treated in some detail in order to exhibit clearly what the approximations are and why the methods work. However, some of this theoretical material can be skipped at first reading (this is the material in the sections marked with an asterisk *). Details on implementation are given for most of the methods described.

We start off with a chapter on quantum mechanical scattering theory. This is a rather straightforward application of numerical techniques, and is used as an

illustration of solving a rather simple (not completely trivial) physical problem on a computer. The results of a sample program are compared with experiment. In [Chapters 3 to 5](#) we discuss computational methods for the electronic structure: variational calculus, Hartree–Fock and density functional theory. We apply these methods to some simple systems: the hydrogen and the helium atoms, and the hydrogen molecule. We calculate the energies of these systems. [Chapter 6](#) deals with solving the independent-particle Schrödinger equation in solids.

In [Chapters 7 to 12](#) we describe molecular dynamics and Monte Carlo techniques for classical and quantum many-particle systems. [Chapter 7](#) contains an overview of classical statistical mechanics, with emphasis on ensembles and on critical phenomena, which are also important for field theory, as discussed in [Chapter 15](#). The molecular dynamics and Monte Carlo techniques are treated in [Chapters 8 and 10](#). The standard example of a molecular liquid, argon, is analysed, but simulations for liquid nitrogen and for lattice spin systems (Ising model) are also discussed. [Chapter 9](#) deals with the quantum molecular dynamics technique.

The relations between classical and statistical mechanics are exploited in [Chapter 11](#) where the transfer matrix method for lattice spin systems is described. The next chapter deals with the application of Monte Carlo methods to quantum mechanics, and we revisit the helium atom which is now treated without Hartree–Fock or DFT approximations.

In [Chapter 15](#) we consider numerical methods for field theory. Techniques for analysing the simplest interesting field theory, the scalar ϕ^4 theory, are studied, and methods for studying more complicated field theories (QED and QCD) are discussed. Because of the relation between statistical and quantum mechanics, some of the techniques discussed in this chapter are also relevant for classical statistical mechanics.

Finally, in [Chapter 16](#) modern computer architectures are briefly considered and an example of a parallel algorithm for molecular dynamics is given.

The algorithms presented, and the programs to be written in the exercises, can be coded in different languages: C, C++, Java, Fortran 77, Fortran 90 etc. Also, an integrated scientific computer environment such as MatLab may be used. They all have their pluses and minuses: Fortran 77 allows for dirty programming, but is quite efficient, and the same holds for C; Fortran 90 is efficient and neat. MatLab is easy to use, but not as efficient as using a high-level programming language. Perhaps the most structured way of programming is by using the objected-oriented programming paradigm, as implemented in the languages C++ and Java. For large and complex projects, these languages are unbeatable. However, for smaller jobs MatLab or Fortran 90 is usually sufficient. It is my experience that students relatively new to programming get their programs to run correctly most quickly when using Fortran 90 or MatLab.

Perhaps the most decisive criterion for choosing a particular language is whether you have experience with it. There is no emphasis on any of these languages in this book: they are all suitable for writing the numerical types of programs considered here. If asked for a recommendation, I would not hesitate to advocate Fortran 90 as the most suitable language for inexperienced programmers. Students with substantial programming skills are probably better off using C++ or Java.

It is hoped that in the future it might be easier to move around pieces of software and embed them in new programs, using graphical user interfaces (GUIs). Object-oriented programming (OOP) techniques will play a major role in this development. Ideally, the programs in this book could be built using a set of building blocks which are graphically connected by the programmer to form programs which include the numerical work along with a user-friendly input/output environment.

Exercises

- 1.1 [C] In this problem it is assumed that you have at your disposal a routine for solving an ordinary second order differential equation with given initial conditions (see also [Appendix A7.1](#)).
- (a) Write a program for the Duffing oscillator [see [Eqs. \(1.1\)](#) and [\(1.2\)](#)] and study the motion for different sets of values of the parameters F_0 , ω , γ (use $m = 1$, $a = 1/4$, $b = 1/2$) and initial conditions x_0 and $\dot{x}_0$.
 - (b) For the same values of the parameters, print out the values of x and $p = \dot{x}$ each time a period $T = 2\pi/\omega$ has elapsed. Plotting these points should yield a structure like [Figure 1.3](#). The resulting curve is called the *strange attractor*.

It is possible to assign a dimension to such a structure. This is done by covering it with grids of different sizes, and counting the number of squares $N(b)$ needed to cover the structure with squares of size $b \times b$. This number then scales with b as

$$N(b) \propto b^{-D_f}, \text{ small } b,$$

and the exponent D_f is then the dimension. This is not always an integer number, and if this is indeed not the case, we call the dimension, and the corresponding structure, *fractal*.

- (c) Argue that the dimension of a line determined in this way is equal to one, and that that of a surface area is equal to two.
- (d) Write a program to determine the fractal dimension of the strange attractor constructed above. This proceeds as follows. First, read all the points of the attractor into an array. The attractor lies in the square of side 6 centred at the origin. Divide this square up into $l \times l$ cells, where l is first taken to be 2, then 4, and so on up to $l = 128$. The side of the cells is then $b = 6/l$. A Boolean array of size at least 128 by 128 assumes the value TRUE for cells which contain a point of the attractor and FALSE if that is not the case. Fill this array by performing a loop over the points on the attractor, and then count the number of filled cells $N(b)$.

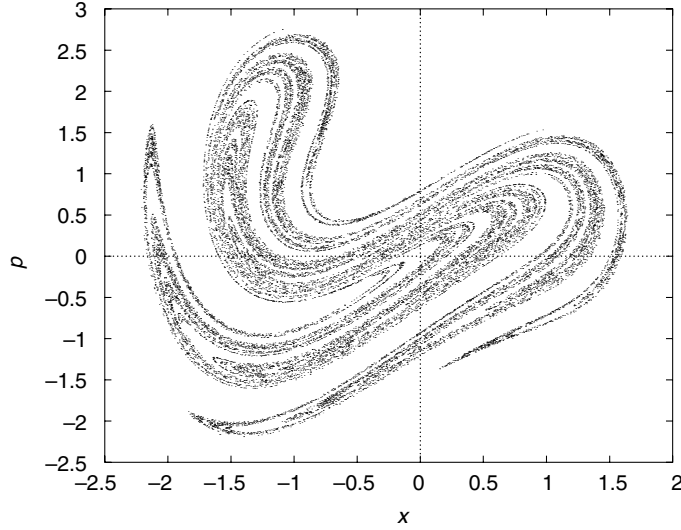


Figure 1.3. Strange attractor for the Duffing oscillator. Values of the parameters are $F_0 = 2.0$, $\omega = 2.4$, $\gamma = 0.1$. The initial conditions are $x_0 = 0.5$, $\dot{x}_0 = 0$.

The results $\log[N(b)]$ and $\log(b)$ should be written to a file. For an attractor of 25 000 points, the resulting points lie more or less on a straight line with slope $-D_f \approx -1.68$, for $2 \leq l \leq 7$.

1.2 [C] In this problem, we consider diffusion limited aggregation.

- (a) Write a program for generating DLA clusters on a square lattice of size 150×150 (see [Section 1.3](#)). Generate a cluster of about 9000 sites, and write the sites occupied by this cluster to a file for viewing using a graphics program.
- (b) Another definition of the fractal dimension (see Problem 1.1) is obtained by relating the number of sites N of the cluster to its *radius of gyration*, defined by

$$R_g = \frac{1}{N} \sum_{i=1}^N (\mathbf{r}_i - \mathbf{r}_0)^2,$$

where

$$\mathbf{r}_0 = \frac{1}{N} \sum_{i=1}^N \mathbf{r}_i$$

is the ‘centre of mass’ of the cluster. Show that the radius of gyration can be rewritten as

$$R_g = \frac{1}{N} \left(\sum_{i=1}^N \mathbf{r}_i^2 \right) - \mathbf{r}_0^2.$$

Use this formula to calculate the radius of gyration after every 200 newly added sites, and write the values $\log(R_g)$, $\log(N)$ to a file. Plot this file and fit the results

to a straight line. The slope of this line is then the fractal dimension which must be about 1.7.

References

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